

# Brandon Downer

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## EDUCATION

### **COOP Careers – Data Analytics Fellow**

*August 2025-December 2025*

- Built applied skills in data cleaning, transformation, and analysis using SQL, Python, and Tableau in business environments.
- Completed 200+ hours working with real-world datasets in collaborative, cross-functional settings.

### **Cornell Tech – Break Through Tech AI Fellow**

*April 2023-May 2024*

- Selected from 1,500+ applicants for an AI/ML program focused on applied ML and real-world system development.
- Improved applied ML model performance through experimentation, validation, and iterative refinement.

### **Lehman College, Bronx, NY**

*Bachelor of Science in Computer Science, Minor in Dance*

*May 2025*

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## WORK EXPERIENCE

### **Caldarium**

**Albany, NY**

*ML & Data Engineer*

*September 2025 – Present*

- Designing and deploying LLM-powered product features using RAG and agentic workflows, integrating AI into real user-facing document processing systems.
- Making technical tradeoffs between prompting, retrieval, and system design approaches to optimize output quality.
- Building and owning end-to-end data pipelines through deployment, ensuring production-grade scalability.
- Implementing evaluation frameworks and guardrails to detect failure modes and reduce hallucinations.

### **Google**

**New York, NY**

*AI/ML Team Lead*

*August 2023 – December 2023*

- Led a team of 5 engineers to design and ship a production ML feature using a convolutional neural network (CNN) and Scikit-Learn-based analysis, improving campaign efficiency by 85%.
- Partnered with stakeholders to define problem scope and determine appropriate ML approaches.
- Built and deployed pipelines integrating models into user-facing workflows, iterating based on real-world feedback.
- Designed evaluation and validation systems to ensure model reliability, robustness, and production readiness.

## PROJECTS

*Caldarium - RPI Medical Parsing Project*

*December 2025*

- Led a cross-functional team, driving development from ambiguity to a deployed solution.
- Applied retrieval-based methods and orchestration logic to improve system accuracy and performance.
- Deployed end-to-end AI systems converting unstructured documents into structured outputs with GenAI pipelines.
- Implemented testing, validation, and performance improvements to ensure production-ready reliability.

*New York Botanical Garden - Plant Identification Machine Learning Project*

*June 2024*

- Developed a computer vision classification model using TensorFlow/Keras for image-based plant identification, covering the full ML lifecycle from data to deployment.
- Improved model performance through iterative experimentation and optimization, achieving 87% accuracy.
- Evaluated model outputs using multiple metrics, refining behavior for consistent real-world performance.

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## TECHNICAL SKILLS & CERTIFICATIONS

- **Certifications:** Google AI Essentials, Machine Learning Foundations
- **Programming & Tools:** Python, Java, SQL, R, React, Tableau
- **AI/ML Systems:** LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), Evaluation
- **MLOps & Data Systems:** API Development, Distributed Data Processing, ML Workflows, Containerization